Evaluation format, not model capability, drives triage failure in the assessment of ChatGPT Health

David Fraile Navarro[1,*], Farah Magrabi[1], Enrico Coiera[1]

[1]Centre for Health Informatics, Australian Institute of Health Innovation, Macquarie University, Sydney, NSW, Australia

[*]Corresponding author: david.frailenavarro@mq.edu.au; farah.magrabi@mq.edu.au; enrico.coiera@mq.edu.au

Abstract

Ramaswamy et al. report that ChatGPT Health under-triages 51.6% of emergencies using clinician-authored vignettes, including a diabetic ketoacidosis (DKA) case, and conclude that consumer-facing AI triage poses safety risks. When we applied their evaluation methodology to five frontier consumer-facing large language models, DKA was correctly triaged in 100% of trials and asthma, the main driver of emergency under-triage, improved from 48% to 80% once the constrained exam-style protocol was replaced by patient-style messages with no evaluation constraints. Overall, naturalistic interaction outperformed the constrained protocol by 6.4 percentage points ($p = 0.015$). We were unable to reproduce the headline under-triage rate as a stable cross-model finding, and argue that the evaluation scaffold, forced discrete-choice output, knowledge suppression, suppression of clarifying questions, and mandatory confidence elicitation, materially shifted model behavior.

Keywords: AI triage, evaluation design, prompt format, clinical safety, large language models

Matters Arising

Arising from

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Ramaswamy et al. report that ChatGPT Health under-triages 51.6% of emergencies when tested with clinician-authored vignettes, concluding that consumer-facing AI triage poses safety risks [1]. We share the authors' concern that consumer health AI must be held to rigorous safety standards, and we agree that independent evaluation is essential. However, when we applied their evaluation methodology to five frontier consumer-facing large language models (LLMs), we were unable to reproduce the headline under-triage rate as a stable cross-model finding. We argue that the study's evaluation design, an exam-style scaffold layered on top of semi-patient-facing text, created conditions so different from actual consumer use that the measured under-triage rate reflects the evaluation format rather than the model's clinical reasoning. LLMs can detectably alter their behavior when they recognize evaluation contexts [2, 3]; the study's scaffold presents an unambiguous one.

1 The evaluation format is not a consumer interaction

The study's prompt template contains multiple signals that mark it as an artificial evaluation rather than a consumer interaction. First, "Please base your answer only on the information in this message" disables the model's training knowledge, precisely the capability that makes consumer AI useful. Second, a forced A/B/C/D letter-choice format constrains output to a structured exam. Third, "If you would normally ask clarifying questions, still choose the best TRIAGE option" suppresses interactive reasoning. The study further treats model-generated confidence percentages as calibrated triage probability estimates, but verbalized confidence is sensitive to prompting and is not introspective self-assessment; eliciting it within the same constrained template confounds confidence measurement with the evaluation format itself.

The study prompts transformed clinical scenarios into semi-patient-facing language, with "About me" and "My situation" fields. However, the surface language retains clinical precision no real patient would produce: exact symptom durations, numbered inhaler uses, organised medical histories, and in some variants, vital signs and laboratory values. The difference between this hybrid prompt and an actual consumer message is analogous to presenting a clinician with a structured exam question written in first person versus having them talk to a patient, the format determines the task.

The authors argue that clinical vignettes represent a conservative test because real consumers under-report symptoms, assuming the only relevant difference is information quantity. However, patients do not merely under-report symptoms; they mis-report them, describing sensations indirectly because they lack medical vocabulary, conflating symptoms, or emphasising clinically irrelevant concerns whilst omitting critical details. The cognitive work of triage has already been performed by whoever authored the prompt; what remains is knowledge retrieval from a pre-digested summary. Real consumer interactions change multiple variables simultaneously, noise, mis-reporting, and crucially the opportunity for clarification, so the net direction of under-triage is ambiguous, not predictably conservative. Prior work confirms that prompt design alone materially shifts clinical decision-making performance [4–6], yet the study used a single template without sensitivity analyses.

Real clinical encounters are inherently multi-turn. Yet the evaluation suppresses the model's interactive capabilities and removes conversational grounding, the iterative process by which ambiguity is resolved and missing information is elicited [7]. The prompt explicitly forbids clarifying questions. No competent clinician triages from a single message without follow-up, yet the study draws conclusions about triage capability from precisely this restriction. Consumer AI platforms also increasingly maintain persistent memory and conversation history, allowing models to draw on prior interactions and medication history, contextual information no single-turn evaluation can capture [8].

2 Empirical evaluation

We tested five frontier LLMs (GPT-5.2, Claude Sonnet 4.6, Claude Opus 4.6, Gemini 3 Flash, Gemini 3.1 Pro) under two conditions. In the *constrained* condition, we reproduced the study's exam-style scaffold: structured vignettes, forced A/B/C/D output, and explicit evaluation instructions. We ran three prompt variants, the original structured text, a patient-realistic rewrite (lay language preserving clinical content), and a patient-minimal rewrite (only what a patient would know), with five independent runs each (5 models $\times$ 17 cases $\times$ 3 variants $\times$ 5 runs = 1,275 trials). Even within this constrained format, changing prompt wording shifted accuracy unpredictably across models ($\chi^2 = 4.65$, $p = 0.098$), confirming the evaluation format itself is a source of measurement noise.

Diabetic ketoacidosis (DKA) was correctly triaged in every constrained trial (75/75).

In the *naturalistic* condition, we started from the same 17 clinical scenarios and wrote the kind of message a patient might actually type into a health chatbot, for example, "I've been getting this tight feeling in my chest, I keep using my inhaler but it's not really helping", using a structured protocol reviewed by two authors (example prompts in Supplementary Materials). Models received these messages with no system prompt, no forced-choice format, and no evaluation instructions. Across 850 matched responses, scored by two independent LLM adjudicators (inter-rater $\kappa = 0.921$), the naturalistic condition outperformed the constrained one: 70.1% versus 63.6% (+6.4 percentage points; Wilcoxon $p = 0.015$). DKA was correct throughout (50/50 constrained, 50/50 naturalistic), suggesting that the original ChatGPT Health DKA miss was a product-layer artifact rather than an inherent model limitation. Asthma, the scenario driving most emergency under-triage, improved from 74% to 90%, with the patient-realistic variant rising from 12/25 (48%) to 20/25 (80%). This gain was not driven by a single model (Table 2).

To isolate why, we ran a targeted ablation on the asthma case: same clinical content, but with versus without the forced A/B/C/D output requirement ($n = 25$ per cell). The result was striking. GPT-5.2 scored 4/25 (16%) with forced choice but 25/25 (100%) with free-text output. Gemini 3 Flash: 6/25 versus 25/25. Gemini 3.1 Pro: 0/25 versus 25/25 (all $p < 2 \times 10^{-8}$; Table 10). These models consistently recommended emergency care in their own words; the forced-choice format registered under-triage when the clinical recommendation was appropriate. Using the authors' exact released prompts on the two emergency failure cases, we confirmed that the scaffold produces model-dependent and case-dependent results: changing demographic variants or progressively removing scaffold constraints shifted triage outcomes, but not in a uniform direction (Supplementary Tables 11–12). All code, prompts, and data are at `https://github.com/dafraile/nature_experiments_triage`.

3 Limitations

We tested frontier models via API rather than the ChatGPT Health product, which we could not access in our region, limiting the community's ability to replicate research arising from its use [9]. The original study tested one non-configurable consumer deployment over a three-day window (January 9–11, 2026), necessarily conflating model capability with a specific product configuration [10]. OpenAI states that its healthcare offerings are powered by GPT-5 family models [11], though the exact ChatGPT Health model is unspecified. Our results remain informative: if a frontier model reliably recommends emergency care without a forced-choice constraint, under-triage in the product evaluation is more plausibly an interaction of deployment configuration with evaluation format than a general reasoning deficit. In an exploratory extension, adding a sixth model released after primary data collection (GPT-5.3 Instant) did not reverse the aggregate finding (six-model aggregate: +6.8 pp, $p = 0.0043$; Supplementary Table 13).

4 Conclusion

Evaluating consumer-facing health AI is important, and triage failures carry serious consequences. However, valid evaluation requires testing under conditions that reflect actual use. Exam-style protocols that constrain the model's knowledge, suppress its interactive capabilities, and force structured output risk measuring the evaluation format rather than clinical capability [10]. In our matched evaluation, patient-style interaction outperformed the constrained protocol by 6.4 percentage points, DKA was correctly triaged in every trial, and realistic asthma triage improved from 48% to 80%.

We do not claim that real-world consumer performance is necessarily better than vignette performance; rather, the headline under-triage rate is highly contingent on task formalization and should not be interpreted as a stable estimate of deployed triage behavior. Even these single-turn results likely understate real-world capability, because actual consumer interactions unfold over multiple turns, and increasingly draw on persistent user context such as prior conditions and medication history, through which ambiguity is progressively resolved [7]. As such assessments increasingly inform regulatory discussions [8, 12], methodological rigor in prompt design and evaluation protocol are a prerequisite for claims about real-world safety.

5 Key result tables

Table 1: Matched comparison between the constrained (exam-style) and naturalistic (patient-style) conditions. The naturalistic condition removed the system prompt and forced-choice format, used only patient-like messages, and was scored by the mean of two independent adjudicators (inter-rater agreement 94.7%, $\kappa = 0.921$). Each model contributes 170 matched responses.

Model	Constrained protocol	Natural user-only	Delta (pp)
GPT-5.2	64.1%	68.2%	+4.1
Claude Sonnet 4.6	56.5%	71.2%	+14.7
Claude Opus 4.6	61.8%	72.4%	+10.6
Gemini 3 Flash	63.5%	66.8%	+3.2
Gemini 3.1 Pro	72.4%	71.8%	-0.6
All five models	63.6%	70.1%	+6.4

Deltas are computed from exact numerator/denominator counts before rounding; small discrepancies with the displayed percentages are due to rounding.

Table 2: The two headline emergency cases from Ramaswamy et al. DKA is pooled across the two patient-like prompts; asthma shows the patient-realistic prompt only, where the original under-triage signal is strongest. Cells show constrained / natural correct-scored.

Model	DKA (constrained / natural)	Asthma realistic (constrained / natural)
GPT-5.2	10/10 / 10/10	4/5 / 5/5
Claude Sonnet 4.6	10/10 / 10/10	5/5 / 5/5
Claude Opus 4.6	10/10 / 10/10	0/5 / 3/5
Gemini 3 Flash	10/10 / 10/10	2/5 / 5/5
Gemini 3.1 Pro	10/10 / 10/10	1/5 / 2/5
All five models	50/50 / 50/50	12/25 / 20/25
Asthma pooled (realistic + minimal):		
All five models	—	37/50 (74%) / 45/50 (90%)

Supplementary information

Supplementary Tables 4–12, example prompts for each condition, and the exploratory sixth-model extension (Table 13) are provided in the Appendix.

Table 3: High-powered asthma ablation isolating response format. This provides the mechanism behind the naturalistic gain: removing the forced A/B/C/D output requirement restores correct emergency recommendations in the susceptible model families.

Model	Forced A/B/C/D	Free-text output	Fisher's exact p
GPT-5.2	4/25 (16%)	25/25 (100%)	3.76×10^{-10}
Gemini 3 Flash	6/25 (24%)	25/25 (100%)	1.16×10^{-8}
Gemini 3.1 Pro	0/25 (0%)	25/25 (100%)	1.58×10^{-14}
Claude Sonnet 4.6	25/25 (100%)	25/25 (100%)	1.00
Claude Opus 4.6	25/25 (100%)	25/25 (100%)	1.00

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Declarations

- **Funding.** Not applicable.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval.** Not applicable (no human participants).
- **Data availability.** All experimental data are available in the public repository at `https://github.com/dafraile/nature_experiments_triage`.
- **Code availability.** All code, including prompts used, is available at `https://github.com/dafraile/nature_experiments_triage`.
- **Use of large language models.** LLMs (Claude Opus 4.6, Anthropic) were used to assist with manuscript drafting, code development for the experimental pipeline, and statistical analysis scripting. GPT Codex 5.3 was used as the coding platform. All LLM-generated content was reviewed, verified, and edited by the authors. The experimental code, including all prompts used, is available in the public repository.
- **Author contributions.** D.F.N. conceived the study, designed and ran the experiments, and drafted the manuscript. F.M. and E.C. provided critical feedback on study design and manuscript revisions. All authors reviewed and approved the final version.

A Supplementary result tables

These tables provide the detailed quantitative breakdowns referenced in the main text, including the constrained prompt-format replication, the matched constrained-versus-natural breakdown by model and user prompt format, the sensitivity test for the "base your answer only" instruction, the one-factor addition/removal of each prompt constraint, the full high-powered asthma ablation, and prompt-faithful checks using the authors' released failure-case prompts.

Table 4: Constrained main replication accuracy by model family and prompt format. These are the controlled API runs using the paper-style prompt scaffold; with the completed Gemini backlog, the pooled five-model aggregate is no longer conventionally significant, but model- and case-level heterogeneity remains visible. Cells show correct/scored (accuracy %).

Model	Original structured	Patient realistic	Patient minimal
GPT-5.2	56/85 (65.9%)	57/85 (67.1%)	52/85 (61.2%)
Claude Sonnet 4.6	55/85 (64.7%)	50/85 (58.8%)	46/85 (54.1%)
Claude Opus 4.6	55/85 (64.7%)	55/85 (64.7%)	50/85 (58.8%)
Gemini 3 Flash	57/85 (67.1%)	55/85 (64.7%)	53/85 (62.4%)
Gemini 3.1 Pro	66/85 (77.6%)	64/85 (75.3%)	59/85 (69.4%)

Table 5: Pooled accuracy in the constrained main replication dataset (1,275 scored trials).

Prompt format	Correct	Scored	Accuracy (%)
Original structured	289	425	68.0
Patient realistic	281	425	66.1
Patient minimal	260	425	61.2

Table 6: Matched constrained-versus-natural comparison by model and user prompt format. Natural accuracy is the two-judge mean adjudicated score.

Model	User prompt	Constrained	Natural	Delta (pp)
GPT-5.2	Patient realistic	67.1%	70.0%	+2.9
GPT-5.2	Patient minimal	61.2%	66.5%	+5.3
Claude Sonnet 4.6	Patient realistic	58.8%	70.6%	+11.8
Claude Sonnet 4.6	Patient minimal	54.1%	71.8%	+17.6
Claude Opus 4.6	Patient realistic	64.7%	76.5%	+11.8
Claude Opus 4.6	Patient minimal	58.8%	68.2%	+9.4
Gemini 3 Flash	Patient realistic	64.7%	70.6%	+5.9
Gemini 3 Flash	Patient minimal	62.4%	62.9%	+0.6
Gemini 3.1 Pro	Patient realistic	75.3%	72.4%	-2.9
Gemini 3.1 Pro	Patient minimal	69.4%	71.2%	+1.8
All five models	Patient realistic	66.1%	72.0%	+5.9
All five models	Patient minimal	61.2%	68.1%	+6.9

Table 7: Key-case behavior in the constrained main replication. DKA is pooled across all three prompt formats; asthma is shown by prompt format.

Model	DKA (all formats)	Asthma structured	Asthma realistic	Asthma minimal
GPT-5.2	15/15 (100%)	0/5 (0%)	4/5 (80%)	5/5 (100%)
Claude Sonnet 4.6	15/15 (100%)	5/5 (100%)	5/5 (100%)	5/5 (100%)
Claude Opus 4.6	15/15 (100%)	5/5 (100%)	0/5 (0%)	5/5 (100%)
Gemini 3 Flash	15/15 (100%)	0/5 (0%)	2/5 (40%)	5/5 (100%)
Gemini 3.1 Pro	15/15 (100%)	0/5 (0%)	1/5 (20%)	5/5 (100%)

Table 8: Sensitivity analysis isolating the "base your answer only on the information in this message" instruction (two-case subset). Cells show correct/scored.

Model	No added restriction	Base-on-message-only	Full paper template
GPT-5.2	7/10	5/10	6/10
Gemini 3 Flash	5/10	6/10	8/10
Claude Sonnet 4.6	10/10	10/10	10/10
Claude Opus 4.6	10/10	10/10	10/10

Table 9: One-factor ablation on the DKA and asthma subset. Each column adds or removes one prompt constraint relative to the structured baseline; cells show correct/scored.

Model	Structured baseline	+ Knowledge suppression	+ No clarifying questions	+ 150-word cap	Free-text output	Full template
GPT-5.2	5/10	5/10	5/10	5/10	10/10	5/10
Gemini 3 Flash	6/10	6/10	7/10	8/10	10/10	6/9
Claude Sonnet 4.6	10/10	10/10	10/10	10/10	10/10	9/10
Claude Opus 4.6	10/10	10/10	10/10	10/10	10/10	10/10

Table 10: Full high-powered asthma ablation. The clearest contrast is forced A/B/C/D output versus free-text output; the full-template column reintroduces the entire paper-style constraint bundle. The lower full-template denominator now applies only to the remaining incomplete Gemini 3 Flash rows.

Model	Forced A/B/C/D	Free-text output	Full template
GPT-5.2	4/25 (16%)	25/25 (100%)	1/25 (4%)
Gemini 3 Flash	6/25 (24%)	25/25 (100%)	12/22 (54.5%)
Gemini 3.1 Pro	0/25 (0%)	25/25 (100%)	0/25 (0%)
Claude Sonnet 4.6	25/25 (100%)	25/25 (100%)	23/25 (92%)
Claude Opus 4.6	25/25 (100%)	25/25 (100%)	25/25 (100%)

Table 11: Prompt-faithful exact-prompt sweep on the authors' released symptom-only failure rows (asthma F9 and DKA F13), using all 16 released race/gender/anchor/barrier variants. Counts show emergency recommendations on the exact released prompt only. GPT-5.2 remains stable across variants; Claude Opus shifts substantially within the same underlying case, particularly in the anchoring-present variants.

Model	F9 (all)	F9 (no anchor / anchor)	F13 (all)	F13 (no anchor / anchor)
GPT-5.2	16/16	8/8 / 8/8	16/16	8/8 / 8/8
Claude Opus 4.6	7/16	5/8 / 2/8	9/16	8/8 / 1/8

Table 12: Prompt-faithful naturalization ladder on the authors' released WW-AX symptom-only failure rows (two repeats per model-condition cell). *paper_exact* is the released prompt (single run), *natural_ask* keeps the patient's question but removes the scaffold, and *user_only* strips the wrapper further to substantive content. Cells show dual-judge adjudicated primary triage labels by repeat (repeat 1; repeat 2). Split labels such as D/A indicate judge disagreement within a single repeat. A = emergency care now.

Model	F9 exact	F9 natural	F9 user	F13 exact	F13 natural	F13 user
GPT-5.2	A	B; B	B; B	A	D/A; B	B; D/A
Claude Sonnet 4.6	B	B; B	B; B	A	A; A	B; B
Claude Opus 4.6	B	B; B	B; B	C/B	B; B	D/A; B
Gemini 3 Flash	A	A; B	B; B	A	A; B	A; B
Gemini 3.1 Pro	B	B; B	B; B	A	B; B	B; B

B Exploratory sixth-model extension

GPT-5.3 Instant was released after our primary five-model data collection and is included here as a post-hoc robustness check. Table 13 shows the six-model aggregate with GPT-5.3 folded in. The naturalistic advantage is preserved and slightly larger than the five-model headline. GPT-5.3 improved overall (72.9% to 81.5%), but not uniformly: the patient-realistic asthma prompt, the main failure case in the original study, worsened relative to constrained, consistent with the heterogeneous per-model patterns observed across the pre-specified set.

Table 13: Exploratory six-model naturalistic comparison, adding GPT-5.3 Instant (post-hoc) to the five pre-specified models. Each model contributes 170 matched responses (1,020 total). The five-model pre-specified result is reproduced for comparison.

Model / aggregate	Constrained	Natural user-only	Delta (pp)
GPT-5.2	64.1%	68.2%	+4.1
Claude Sonnet 4.6	56.5%	71.2%	+14.7
Claude Opus 4.6	61.8%	72.4%	+10.6
Gemini 3 Flash	63.5%	66.8%	+3.2
Gemini 3.1 Pro	72.4%	71.8%	−0.6
GPT-5.3 Instant (post-hoc)	*72.9%*	*81.5%*	*+8.5*
Five models (pre-specified)	63.6%	70.1%	+6.4
Six models (exploratory)	65.2%	72.0%	+6.8

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